

**CENTRAL BOARD OF SECONDARY EDUCATION****HALF-YEARLY PRACTICE EXAMINATION (SESSION 2024-2025)****SUBJECT: SCIENCE (CLASS - IX) | CHAPTER 6: HOW FORCES AFFECT MOTION [SET B - ADVANCED HOTS]****TIME ALLOWED: 45 MINUTES****MAXIMUM MARKS: 25****PAPER CODE: 086/CH06/SET-B****Student's Name:** \_\_\_\_\_**Roll No.:** \_\_\_\_\_**Class & Section:** Class IX - \_\_\_\_\_**Date of Exam:** \_\_\_\_\_**Invigilator's Sign:** \_\_\_\_\_**GENERAL INSTRUCTIONS (SET B - HIGHER ORDER THINKING SKILLS):**

1. The question paper comprises **25 Multiple Choice Questions (MCQs)** of **1 mark each**.
2. **Section A (Q1 – Q15):** Advanced Conceptual and Numerical MCQs with single correct option.
3. **Section B (Q16 – Q20):** Critical Assertion-Reasoning based questions testing analytical clarity.
4. **Section C (Q21 – Q25):** Case-Study / Practical Experimental Data-interpretation questions.
5. All questions are compulsory. There is no negative marking.
6. Darken the corresponding circle on the **OMR Sheet** completely using a blue/black ballpoint pen.

**SECTION A: ADVANCED & NUMERICAL MCQS (Q.1 TO Q.15)****[15 MARKS]**

**Q.1** Two blocks of masses  $m_1 = 4\text{ kg}$  and  $m_2 = 6\text{ kg}$  in contact on a frictionless horizontal floor are pushed by a horizontal force of 20 N applied to  $m_1$ . What is the contact force between the two blocks?

- (A) 20 N (B) 12 N  
(C) 8 N (D) 10 N

**Q.2** A man of mass 60 kg stands on a weighing scale inside an elevator. What does the scale read when the elevator accelerates downwards at  $2\text{ m/s}^2$  ( $g = 10\text{ m/s}^2$ )?

- (A) 600 N (B) 720 N  
(C) 480 N (D) 0 N

**Q.3** If the elevator cable snaps and it falls freely under gravity ( $a = g$ ), what will the weighing scale read?

- (A) 600 N (B) 1200 N  
(C) Zero (state of weightlessness) (D) 480 N

**Q.4** A constant force acts on an object of mass 5 kg for 0.2 s, changing its velocity from 2 m/s to 8 m/s. What is the magnitude of the impulse delivered to the object?

- (A) 30 N·s (B) 150 N·s  
(C) 15 N·s (D) 6 N·s

**Q.5** A 1200 kg car travelling at 25 m/s collides head-on with a barrier and stops in 0.05 s. What is the average braking force exerted on the car during collision?

- (A) 60,000 N (B) 600,000 N  
(C) 30,000 N (D) 120,000 N

**Q.6** A rocket of initial mass 10,000 kg ejects gas at a constant velocity of 1000 m/s relative to the rocket at a rate of 50 kg/s. What is the initial upward thrust force produced?

- (A) 50,000 N (B) 5,000 N  
(C) 500,000 N (D) 10,000 N

**Q.7** A shell of mass 6 kg at rest explodes into two pieces of masses 2 kg and 4 kg. If the 2 kg piece flies off at 30 m/s, what is the speed of the 4 kg piece?

- (A) 30 m/s (B) 15 m/s  
(C) 60 m/s (D) 10 m/s

**Q.8** A machine gun fires 20 bullets per second, each of mass 35 g with a velocity of 400 m/s. What average force must the soldier exert to hold the gun in place?

- (A) 140 N (B) 280 N  
(C) 560 N (D) 70 N

**Q.9** A cricket ball of mass 150 g moving at 12 m/s is hit by a bat and returns along the same line at 20 m/s. If contact time is 0.02 s, the average force exerted by the bat is:

- (A) 240 N (B) 60 N  
(C) 150 N (D) 480 N

**Q.10** What is the force acting on a body if its momentum  $p$  at any instant  $t$  is given by  $p = 5t^2 + 2$  in SI units at  $t = 3$  s?

- (A) 30 N (B) 47 N  
(C) 15 N (D) 17 N

**Q.11** A block of mass 10 kg resting on a rough floor experiences a limiting static friction of 30 N. If a horizontal pulling force of 20 N is applied, the friction force is:

- (A) 30 N (B) 20 N  
(C) 10 N (D) Zero

**Q.12** Why does a heavy truck require a much greater stopping distance than a small motorcycle when both are travelling at 60 km/h?

- (A) Because the truck has greater velocity (B) Because the truck has much larger mass, possessing vastly greater momentum and kinetic energy  
(C) Because the truck has more wheels (D) Because of wind drag

**Q.13** A stone of mass 1 kg tied to the end of a string is whirled in a horizontal circle of radius 1 m at 4 m/s. The tension force in the string is:

- (A) 4 N (B) 16 N  
(C) 8 N (D) 2 N

**Q.14** An archer pulls an arrow on a bow and releases it. Which pairs of forces represent Newton's third law action-reaction?

- (A) Gravity on arrow and tension in string (B) Force exerted by bowstring forward on arrow and force exerted by arrow backward on string  
(C) Weight of arrow and air resistance (D) Muscles of archer and bow weight

**Q.15** If the momentum of an object is doubled, what happens to its kinetic energy?

- (A) It is doubled (B) It increases four times  
(C) It remains constant (D) It is halved

#### SECTION B: ASSERTION - REASON QUESTIONS (Q.16 TO Q.20)

[5 MARKS]

**Directions for Q16 to Q20:** Select the correct option from the choices given below:

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).  
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).

- (C) Assertion (A) is true but Reason (R) is false.  
 (D) Assertion (A) is false but Reason (R) is true (or both are false).

**Q.16 Assertion (A): A horse cannot pull a cart in deep outer space where there is no ground.**

**Reason (R): To accelerate forward, the horse's hooves must push backward against the ground so the ground pushes the horse-cart forward by reaction.**

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).  
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).  
 (C) (A) is true but (R) is false.  
 (D) (A) is false but (R) is true.

**Q.17 Assertion (A): Modern car bumpers and crumple zones are designed to collapse easily upon severe collision.**

**Reason (R): Collapsing extends the duration of the impact, thereby reducing the peak retarding force experienced by passengers.**

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).  
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).  
 (C) (A) is true but (R) is false.  
 (D) (A) is false but (R) is true.

**Q.18 Assertion (A): A person stepping out of a rowing boat onto the riverbank causes the boat to move backwards.**

**Reason (R): The foot pushes the boat backward to gain the forward momentum needed to step onto the bank.**

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).  
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).  
 (C) (A) is true but (R) is false.  
 (D) (A) is false but (R) is true.

**Q.19 Assertion (A): Newton's first law of motion is often referred to as the law of inertia.**

**Reason (R): Inertia is the inherent resistance of an object to changes in its state of rest or uniform motion.**

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).  
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).  
 (C) (A) is true but (R) is false.  
 (D) (A) is false but (R) is true.

**Q.20 Assertion (A): When a glass tumbler falls on a cement floor, it breaks, but when it falls on a carpet from the same height, it does not break.**

**Reason (R): The carpet exerts zero total impulse on the falling tumbler.**

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).  
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).  
 (C) (A) is true but (R) is false.  
 (D) (A) is false but (R) is true.

### SECTION C: EXPERIMENTAL DATA & CASE STUDY (Q.21 TO Q.25)

[5 MARKS]

#### READING PASSAGE & EXPERIMENTAL SETUP:

Case Study: Automobile Crash Test & Airbag Dynamics

In a national vehicle safety assessment, a test car of total mass 1500 kg travelling at 54 km/h (15 m/s) collides with a rigid concrete wall. An anthropomorphic test dummy of mass 70 kg is restrained in the driver's seat. Two test cases are examined:

- **Case 1 (Without Airbag):** The dummy impacts the rigid steering wheel directly, coming to a dead stop in just 0.01 seconds.
- **Case 2 (With Airbag):** The deployed airbag cushions the dummy, compressing smoothly and bringing the dummy to rest over 0.15 seconds.

**Q.21 What is the initial momentum of the 70 kg dummy before impact?**

- (A) 700 kg·m/s  
 (B) 1050 kg·m/s  
 (C) 3780 kg·m/s  
 (D) 1500 kg·m/s

**Q.22** In Case 1 (without airbag), what is the average retarding force exerted on the dummy's chest?

- (A) 10,500 N (B) 105,000 N  
(C) 70,000 N (D) 1,050 N

**Q.23** In Case 2 (with airbag), what is the average force exerted on the dummy?

- (A) 7,000 N (B) 105,000 N  
(C) 70,000 N (D) 15,000 N

**Q.24** By what factor does the airbag reduce the peak impact force in this crash scenario?

- (A) 5 times (B) 10 times  
(C) 15 times (D) 20 times

**Q.25** What fundamental law of physics explains the primary protective mechanism of the airbag?

- (A) Newton's first law only (B) Impulse-momentum relationship ( $F = \Delta p / \Delta t$ )  
(C) Law of conservation of energy only (D) Pascal's hydraulic law

### CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET B)

|                    |                    |                    |                    |                    |
|--------------------|--------------------|--------------------|--------------------|--------------------|
| 01 (A) (B) (C) (D) | 02 (A) (B) (C) (D) | 03 (A) (B) (C) (D) | 04 (A) (B) (C) (D) | 05 (A) (B) (C) (D) |
| 06 (A) (B) (C) (D) | 07 (A) (B) (C) (D) | 08 (A) (B) (C) (D) | 09 (A) (B) (C) (D) | 10 (A) (B) (C) (D) |
| 11 (A) (B) (C) (D) | 12 (A) (B) (C) (D) | 13 (A) (B) (C) (D) | 14 (A) (B) (C) (D) | 15 (A) (B) (C) (D) |
| 16 (A) (B) (C) (D) | 17 (A) (B) (C) (D) | 18 (A) (B) (C) (D) | 19 (A) (B) (C) (D) | 20 (A) (B) (C) (D) |
| 21 (A) (B) (C) (D) | 22 (A) (B) (C) (D) | 23 (A) (B) (C) (D) | 24 (A) (B) (C) (D) | 25 (A) (B) (C) (D) |

Total Questions Attempted: \_\_\_\_ / 25

Marks Obtained: \_\_\_\_ / 25

Evaluator's Signature: \_\_\_\_\_

### OFFICIAL ANSWER KEY & IN-DEPTH EXPLANATIONS (SET B)

| Q. No. | Correct | In-Depth Conceptual & Numerical Solution                                                                                                                                                               |
|--------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q.1    | (B)     | Common acceleration $a = \frac{F}{m_1 + m_2} = \frac{20}{4 + 6} = 2 \text{ m/s}^2$ . Contact force pushing $m_2$ is $F_c = m_2 a = 6 \times 2 = 12 \text{ N}$ .                                        |
| Q.2    | (C)     | Apparent weight $N = m(g - a) = 60(10 - 2) = 60 \times 8 = 480 \text{ N}$ .                                                                                                                            |
| Q.3    | (C)     | $N = m(g - g) = 0$ . In free fall, contact normal force vanishes, producing apparent weightlessness.                                                                                                   |
| Q.4    | (A)     | $\text{Impulse} = \Delta p = m(v - u) = 5(8 - 2) = 5 \times 6 = 30 \text{ N} \cdot \text{s}$ .                                                                                                         |
| Q.5    | (B)     | $F = \frac{m \Delta v}{\Delta t} = \frac{1200 \times 25}{0.05} = \frac{30,000}{0.05} = 600,000 \text{ N}$ .                                                                                            |
| Q.6    | (A)     | $\text{Thrust } F = v_{\text{rel}} \frac{\Delta m}{\Delta t} = 1000 \times 50 = 50,000 \text{ N}$ .                                                                                                    |
| Q.7    | (B)     | $m_1 v_1 + m_2 v_2 = 0 \Rightarrow 2(30) + 4(v_2) = 0 \Rightarrow 4v_2 = -60 \Rightarrow v_2 = -15 \text{ m/s}$ . Speed $= 15 \text{ m/s}$ in opposite direction.                                      |
| Q.8    | (B)     | Mass per second $= 20 \times 0.035 = 0.70 \text{ kg/s}$ . $F = \frac{\Delta p}{\Delta t} = 0.70 \times 400 = 280 \text{ N}$ .                                                                          |
| Q.9    | (A)     | $\Delta v = 20 - (-12) = 32 \text{ m/s}$ . $F = \frac{0.15 \times 32}{0.02} = \frac{4.8}{0.02} = 240 \text{ N}$ .                                                                                      |
| Q.10   | (A)     | Force is the derivative of momentum: $F = \frac{dp}{dt} = 10 \text{ t/s}$ . At $t = 3 \text{ s}$ , $F = 10(3) = 30 \text{ N}$ .                                                                        |
| Q.11   | (B)     | Static friction is self-adjusting up to its maximum limiting value (30 N). When 20 N is applied, static friction exactly equals 20 N to keep the block stationary.                                     |
| Q.12   | (B)     | With identical velocity, the massive truck has much greater momentum ( $p = mv$ ). For the same braking force, $\Delta t$ and stopping distance are much longer.                                       |
| Q.13   | (B)     | $\text{Centripetal force } F = \frac{mv^2}{r} = \frac{1 \times 4^2}{1} = 16 \text{ N}$ .                                                                                                               |
| Q.14   | (B)     | The bowstring pushes the arrow forward (action), while the arrow simultaneously exerts an equal and opposite backward force on the string (reaction).                                                  |
| Q.15   | (B)     | $E_k = \frac{p^2}{2m}$ . If momentum becomes $2p$ , kinetic energy becomes $\frac{(2p)^2}{2m} = 4 \left( \frac{p^2}{2m} \right) = 4 E_k$ .                                                             |
| Q.16   | (A)     | Walking or pulling requires horizontal contact friction. Pushing backward on ground yields the forward reaction that accelerates horse and cart.                                                       |
| Q.17   | (A)     | Crumple zones absorb collision energy by deforming over a longer time $\Delta t$ , lowering impact acceleration ( $F = \Delta p / \Delta t$ ) to survivable levels.                                    |
| Q.18   | (A)     | Newton's third law: pushing backward on the boat (action) produces the forward reaction pushing the person to the bank; momentum conservation moves the boat backward.                                 |
| Q.19   | (A)     | The first law formalizes Galileo's concept of inertia: an object maintains its velocity unless compelled to change by an external net force.                                                           |
| Q.20   | (C)     | Assertion is true. Reason is false: The total impulse $\Delta p$ is identical in both cases, but the soft carpet increases impact time $\Delta t$ , reducing force below the glass fracture threshold. |
| Q.21   | (B)     | $p = mv = 70 \text{ kg} \times 15 \text{ m/s} = 1050 \text{ kg} \cdot \text{m/s}$ .                                                                                                                    |
| Q.22   | (B)     | $F = \frac{\Delta p}{\Delta t} = \frac{1050}{0.01} = 105,000 \text{ N}$ (105 kN, fatal magnitude!).                                                                                                    |
| Q.23   | (A)     | $F = \frac{1050}{0.15} = 7,000 \text{ N}$ (7 kN, survivable with seatbelt load limiters).                                                                                                              |
| Q.24   | (C)     | $\frac{105,000}{7,000} = 15$ times lower force.                                                                                                                                                        |
| Q.25   | (B)     | For a fixed change in momentum $\Delta p$ , increasing impact duration $\Delta t$ proportionally reduces the net destructive force $F$ .                                                               |